Stats 21: Homework 1

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Acknowledgements: several of these problems are copied from or modified from Think Python by Allen Downey.

I've started the homework file for you. You'll need to fill in the rest with your answers.

My encouragement is to use the keyboard shortcuts as much as possible and use the mouse as little as possible while working the Jupyter Notebook.

After you complete the homework with your answers, in the toolbar, select "Clear All Outputs" and then click "Run All".

Review the document to **make sure all requested output is visible**. You will not get credit for problems where the requested output is not visible, even if the function you coded is correct.

When you are satisfied with the output, click the `...` (to the right of Outline), select "Export", and then Export to HTML.Finally, open the HTML file and "Print as PDF". Submit the PDF to Gradescope.

Submit this ipynb file, complete with your answers and output to Canvas / Bruin Learn.

Again, you must make sure all requested output is visible to receive full credit.

Task 1

Create an account on GitHub.

Change your profile picture. Ideally, use photo of yourself that would be appropriate for a resume. If you are not comfortable with the idea of using a photo of yourself, use any other image that is suitable for a workplace environment.

Follow the instructions provided in class to fork the class repository to your GitHub.

Create another repository with at least two text files in it on GitHub (other than the forked class notes repository). Make at least two additional commits to the reposiotry and push them to GitHub.

Provide a link to both repositories here.

You will also need to submit the link to your own repository (not the forked one) to Canvas / Bruin Learn.

Your Answer:

- Link to your forked repository: https://github.com/buikeith112/26-spring-stats-21
- Link to your own repository: https://github.com/buikeith112/test-repository

Problem 2

An important part of programming is learning to interpret error messages and understanding what correction needs to be made.

Read and familiarize yourself with the following error messages.

Explain the error. Then duplicate each cell and correct the error. The first problem has been done for you as an example.

In [32]:
```
# A
print("Hello World"
```

```
  Cell In[32], line 2
    print("Hello World"
                       ^
SyntaxError: incomplete input
```

Answer: The `print()` function is missing the closing parenthesis. This results in an unexpected EOF error.

In [33]:
```
# corrected:
print("Hello World")
```

```
Hello World
```

In [34]:
```
# B
print("Hello")
    print("Goodbye")
```

```
  Cell In[34], line 3
    print("Goodbye")
    ^
IndentationError: unexpected indent
```

Answer: This is an indentation error where there is an indented statement that is not after an if statement or a for loop. This statement should not be indented and raises an error.

In [35]:
```
# Corrected
print("Hello")
print("Goodbye")
```

```
Hello
Goodbye
```

In [36]:
```python
# C
x = 10
if x > 8
    print("x is greater than 8")
```

```
  Cell In[36], line 3
    if x > 8
            ^
SyntaxError: expected ':'
```

Answer: the colon after `if x > 8` is missing. If statements in Python should be immediately followed by a colon to create a block of code that belongs to the if statement. These following statements must be indented.

In [37]:
```python
# Corrected
x = 10
if x > 8:
    print("x is greater than 8")
```

```
x is greater than 8
```

In [38]:
```python
# D
if x = 10:
    print("x is equal to 10")
```

```
  Cell In[38], line 2
    if x = 10:
       ^
SyntaxError: invalid syntax. Maybe you meant '==' or ':=' instead of '='?
```

Answer: The error comes from the statement `if x = 10` because it uses the single = sign operator which represents assignment such as when assigning values to variables. The statement is trying to test for equality like if x is equal to 10, then do this. For these kinds of statements, the double == sign operator should be used.

In [39]:
```python
# Corrected
if x == 10:
    print("x is equal to 10")
```

```
x is equal to 10
```

In [40]:
```python
# E
x = 5
if x == 5:
print("x is five")
```

```
  Cell In[40], line 4
    print("x is five")
    ^
IndentationError: expected an indented block after 'if' statement on line 3
```

Answer: Since the print statement follows the if statement with a colon, it should be indented because it belongs to that if statement's code block.

In [41]:
```
# Corrected
x = 5
if x == 5:
  print("x is five")
```

```
x is five
```

In [42]:
```
# F
l = [1, 2, 50, 10]
l = sort(l)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[42], line 3
      1 # F
      2 l = [1, 2, 50, 10]
----> 3 l = sort(l)

NameError: name 'sort' is not defined
```

Answer: The `sort()` function in Python works directly modifies the object that is using it, but does not return a new value so it cannot be assigned to a new variable. in python sort works in place and it modifies directly so it doesnt return a new sorted list so we cant assign it to a variable

In [43]:
```
# Corrected
l = [1, 2, 50, 10]
l.sort()
print(l)
```

```
[1, 2, 10, 50]
```

Problem 3

Use Python as a calculator. Enter the appropriate calculation in a cell and be sure the output value is visible.

A. How many seconds are there in 42 minutes 42 seconds?

In [44]:
```
print(42*60 + 42)
```

```
2562
```

B. There are 1.61 kilometers in a mile. How many miles are there in 10 kilometers?

In [45]:
```
print(10 / 1.61)
```

```
6.211180124223602
```

C. If you run a 10 kilometer race in 42 minutes 42 seconds, what is your average 1-mile pace (time to complete 1 mile in minutes and seconds)? What is your average speed in miles per hour?

In [46]:
```
# convert kilometers to miles
miles = 10 / 1.61

# get total time in seconds
seconds = 42 * 60 + 42

# Calculate average 1-mile pace in minutes and seconds
seconds_per_mile = seconds / miles
avg_pace_minutes = int(seconds_per_mile // 60)
avg_pace_seconds = int(seconds_per_mile % 60)
print(f"Average pace: {avg_pace_minutes}:{avg_pace_seconds:02}")

# calculate miles per hour
mph = miles / (seconds / 3600)
print(f"mph: {mph}")
```

```
Average pace: 6:52
mph: 8.727653570337614
```

Problem 4

Write functions for the following problems.

A. The volume of a sphere with radius r is

$$V = \frac{4}{3}\pi r^3$$

Write a function `sphere_volume(r)` that will accept a radius as an argument and return the volume.

- Use the function to find the volume of a sphere with radius 5.
- Use the function to find the volume of a sphere with radius 15.

In [47]:
```
import math

def sphere_volume(r):
    volume = (4/3) * math.pi * r**3
    return volume

print(sphere_volume(5))
print(sphere_volume(15))
```

```
523.5987755982989
14137.166941154068
```

B. Suppose the cover price of a book is $24.95, but bookstores get a 40% discount. Shipping costs $3 for the first copy and 75 cents for each additional copy.

Write a function `wholesale_cost(books)` that accepts an argument for the number of books and will return the total cost of the books plus shipping.

- Use the function to find the total wholesale cost for 60 copies.
- Use the function to find the total wholesale cost for 10 copies.

In [48]:
```
def wholesale_cost(books):
        cover_price = 24.95
        discount = 0.4
        # books - 1 because the first book is flat $3 shipping, and the rest are $0
        shipping_cost = 3 + (0.75 * (books - 1))
        total_cost = (cover_price * (1 - discount) * books) + shipping_cost
        # round to nearest cent
        return round(total_cost, 2)

print(wholesale_cost(60))
print(wholesale_cost(10))
```

```
945.45
159.45
```

C. A person runs several miles. The first and last miles are run at an 'easy' pace. Other than the first and last miles, the other miles are at a faster pace.

Write a function `run_time(miles, warm_pace, fast_pace)` to calculate the time the runner will take. The function accepts three input arguments: how many miles the runner travels (minimum value is 2), the warm-up and cool-down pace, the fast pace. The function will print the time in the format minutes:seconds, and will return a tuple of values: (minutes, seconds)

Use the function to find the time to run a total of 5 miles. The warm-up pace is 8:15 per mile. The speed pace is 7:12 per mile.

Call the function using: `run_time(miles = 5, warm_pace = 495, fast_pace = 432)`

In [49]:
```
def run_time(miles, warm_pace, fast_pace):
    # first and last mile have warm pace, the rest have fast pace (miles - 2)
    warm_miles = warm_pace * 2
    fast_miles = fast_pace * (miles - 2)
    # total time in seconds
    total_time = warm_miles + fast_miles
    # convert total time to minutes and seconds
    minutes = total_time // 60
    seconds = total_time % 60
    print(f"{minutes}:{seconds:02}")
    return (minutes, seconds)

print(run_time(miles = 5, warm_pace = 495, fast_pace = 432))
```

```
38:06
(38, 6)
```

Another important skill is to be able to read documentation.

Read the documentation for the function str.split() at https://docs.python.org/3/library/stdtypes.html#str.split

Adjust the function so that the call can be made with minutes and seconds:

`run_time(miles = 5, warm_pace = "8:15", fast_pace = "7:12")`

In [50]:
```python
def run_time(miles, warm_pace, fast_pace):
    # if pace is given with minutes and seconds as a string, convert to list of min
    if type(warm_pace) == str and type(fast_pace) == str:
        # str.split() splits string into a list of elements split at ":" -> (minute
        warm_pace = str.split(warm_pace, ":")
        fast_pace = str.split(fast_pace, ":")
    # if pace is given in seconds, convert to tuple of (minutes, seconds)
    else:
        warm_pace = (warm_pace // 60, warm_pace % 60)
        fast_pace = (fast_pace // 60, fast_pace % 60)
    # extract minutes and seconds from pace and cast to int, then convert to total
    warm_miles = (int(warm_pace[0]) * 60 + int(warm_pace[1])) * 2
    fast_miles = (int(fast_pace[0]) * 60 + int(fast_pace[1])) * (miles - 2)
    # total time in seconds
    total_time = warm_miles + fast_miles
    # convert to minutes and seconds
    minutes = total_time // 60
    seconds = total_time % 60
    print(f"{minutes}:{seconds:02}")
    return (minutes, seconds)

# should work for both string and integer inputs for pace
print(run_time(miles = 5, warm_pace = "8:15", fast_pace = "7:12"))
print(run_time(miles = 5, warm_pace = 495, fast_pace = 432))
```

```
38:06
(38, 6)
38:06
(38, 6)
```

Problem 5

Use `import math` to gain access to the math library.

Create a function `polar(real, imaginary)` that will return the polar coordinates of a complex number.

The input arguments are the real and imaginary components of a complex number. The function will return a tuple of values: the value of the radius `r` and the angle `theta`.

For a refresher, see: https://ptolemy.berkeley.edu/eecs20/sidebars/complex/polar.html

Show the results for the following complex numbers:

- 1 + i
- -2 - 3i
- 4 + 2i

In [51]:
```
import math
def polar(real, imaginary):
    # r^2 = real^2 + imaginary^2
    r = math.sqrt(real**2 + imaginary**2)
    # theta = arctan(imaginary / real)
    # atan2 gives the correct polar coordinates for all quadrants
    theta = math.atan2(imaginary, real)
    return (r, theta)

print(polar(1, 1))
print(polar(-2, -3))
print(polar(4, 2))
```

```
(1.4142135623730951, 0.7853981633974483)
(3.605551275463989, -2.158798930342464)
(4.47213595499958, 0.4636476090008061)
```

Problem 6

Define a function called `insert_into(listname, index, iterable)`. It will accept three arguments, a currently existing list, an index, and another list/tuple that will be inserted at the index position.

Python's built-in function, `list.insert()` can only insert one object.

In [52]:
```
# write your code here
def insert_into(listname, index, iterable):
    for element in iterable:
        listname.insert(index, element)
        index += 1
    return listname
```

In [53]:
```
# do not modify. We will check this result for grading
l = [0,'a','b','c',4,5,6]
i = ['hello', 'there']
insert_into(l, 3, i)
```

Out[53]:
```
[0, 'a', 'b', 'hello', 'there', 'c', 4, 5, 6]
```

Problem 7

Define a function called `first_equals_last(listname)`

It will accept a list as an argument. It will return `True` if the first and last elements are equal and the if the list has a length greater than 1. It will return False for all other cases.

In [54]:
```
# write your function here

def first_equals_last(listname):
    if listname[0] == listname[-1]:
        return True
    return False
```

In [55]:
```
# do not modify. We will check this result for grading
a = [1,2,3]
first_equals_last(a)
```

Out[55]:
```
False
```

In [56]:
```
# do not modify. We will check this result for grading
b = ['hello','goodbye','hello']
first_equals_last(b)
```

Out[56]:
```
True
```

In [57]:
```
# do not modify. We will check this result for grading
c  = [1,2,3,'1']
first_equals_last(c)
```

Out[57]:
```
False
```

In [58]:
```
# do not modify. We will check this result for grading
d = [[1,2],[3,2],[1,2]]
first_equals_last(d)
```

Out[58]:
```
True
```